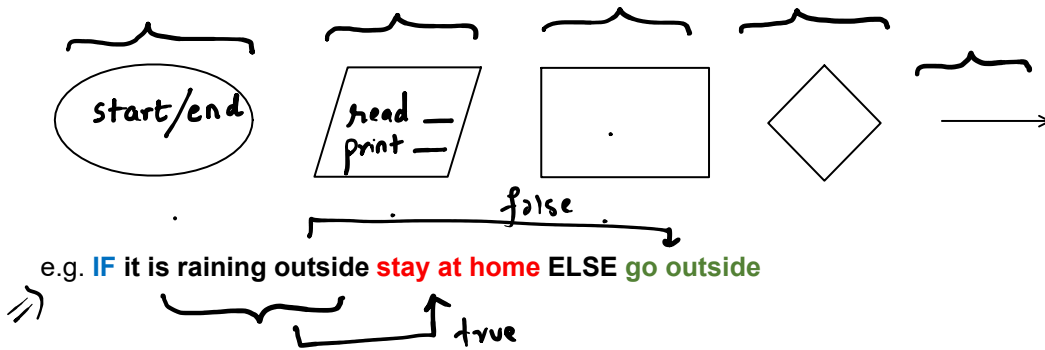


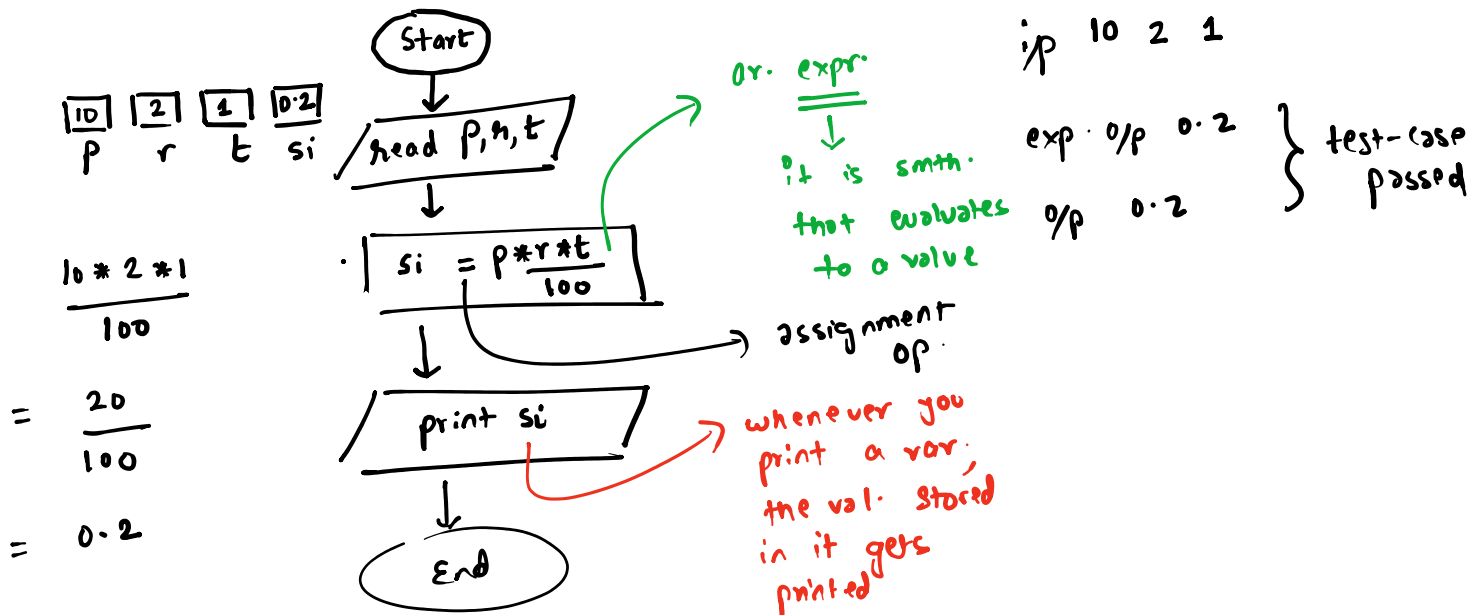
Flow-Chart

It is a **diagrammatic** way of expressing an **algorithm**.



Simple Interest

Given the values of **principal**, **rate** and **time**, design an algorithm to compute the **simple interest** and express it by means of a flowchart.



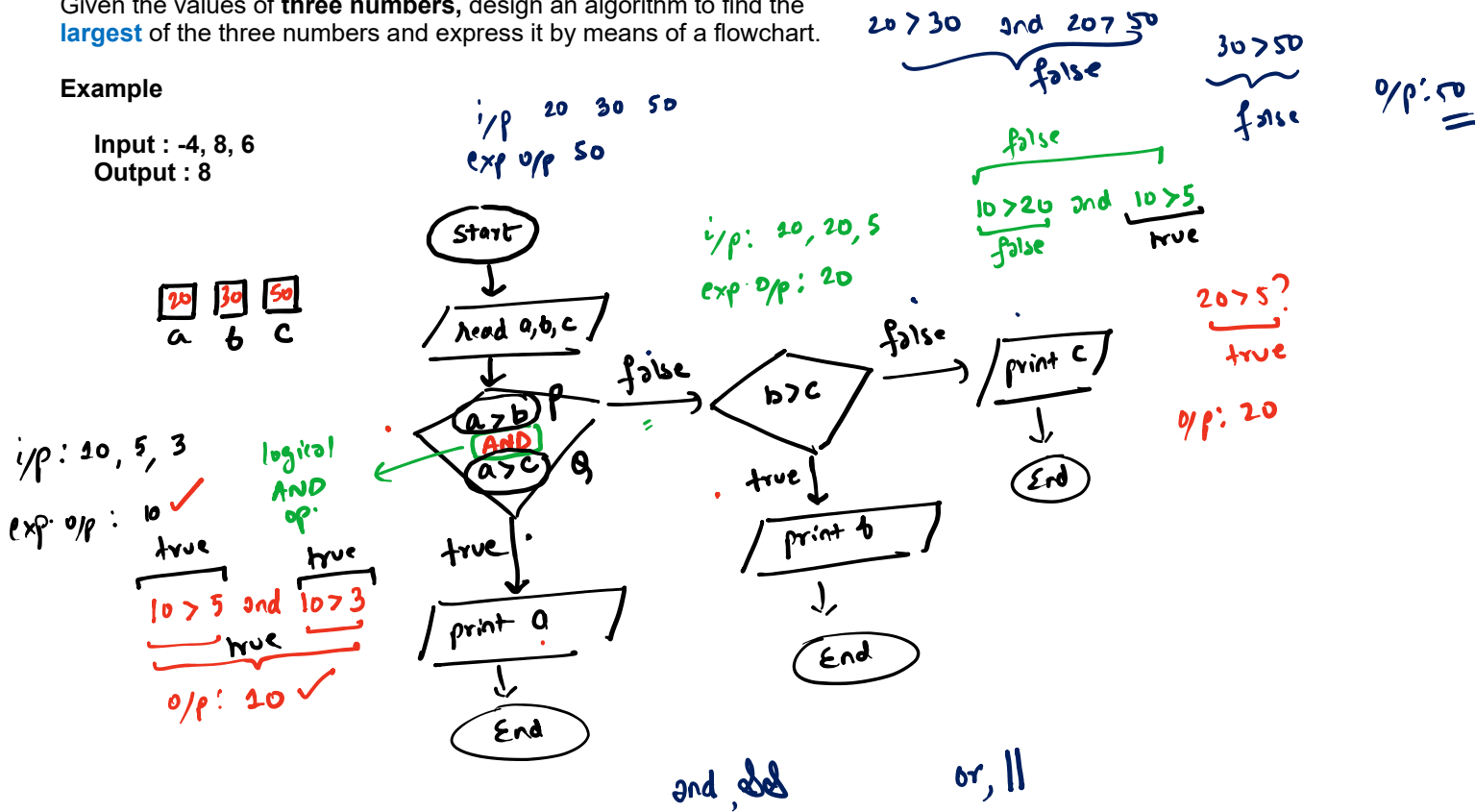
Largest of three numbers

(int)

Given the values of **three numbers**, design an algorithm to find the **largest** of the three numbers and express it by means of a flowchart.

Example

Input : -4, 8, 6
Output : 8



P
True
True
False
False

Q
True
False
True
False

P and Q
True
False
False
False

P or Q
True
True
True
False

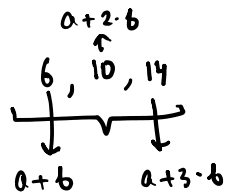
P not P
true false
false true

Print Series

Given two numbers say 'a' and 'b', design an algorithm to **print** the following **series** - $[a+b, a+2 \cdot b, a+3 \cdot b]$ & express it by a flowchart.

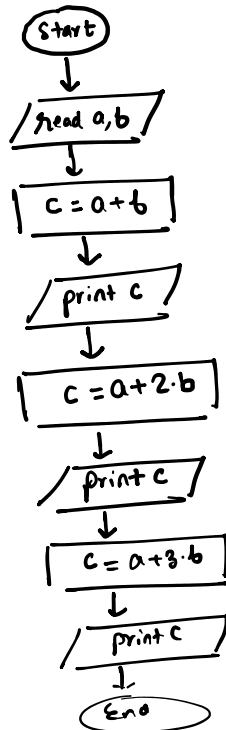
Example

Input : $a=2, b=4$
Output : 6, 10, 14



$\begin{matrix} \square & \square \\ a & b \end{matrix}$

$$\begin{aligned} c &= a + 1 \cdot b \\ c &= a + 2 \cdot b \\ c &= a + 3 \cdot b \\ \hline c &= a + i \cdot b \\ 1 \leq i \leq 3 \end{aligned}$$

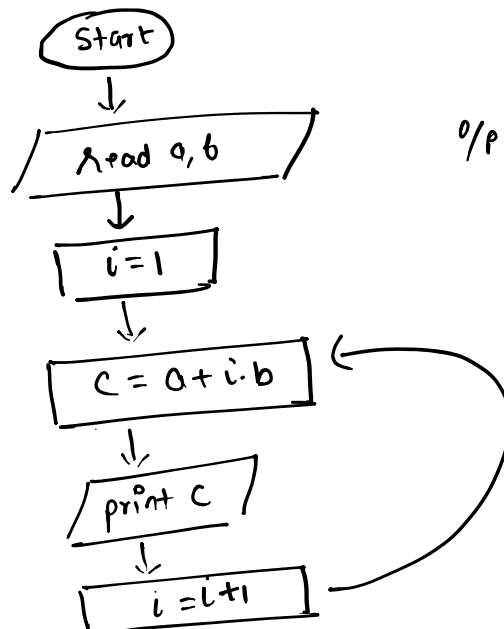


$$\begin{aligned} x &= a + b \\ y &= a + 2 \cdot b \\ z &= a + 3 \cdot b \end{aligned}$$

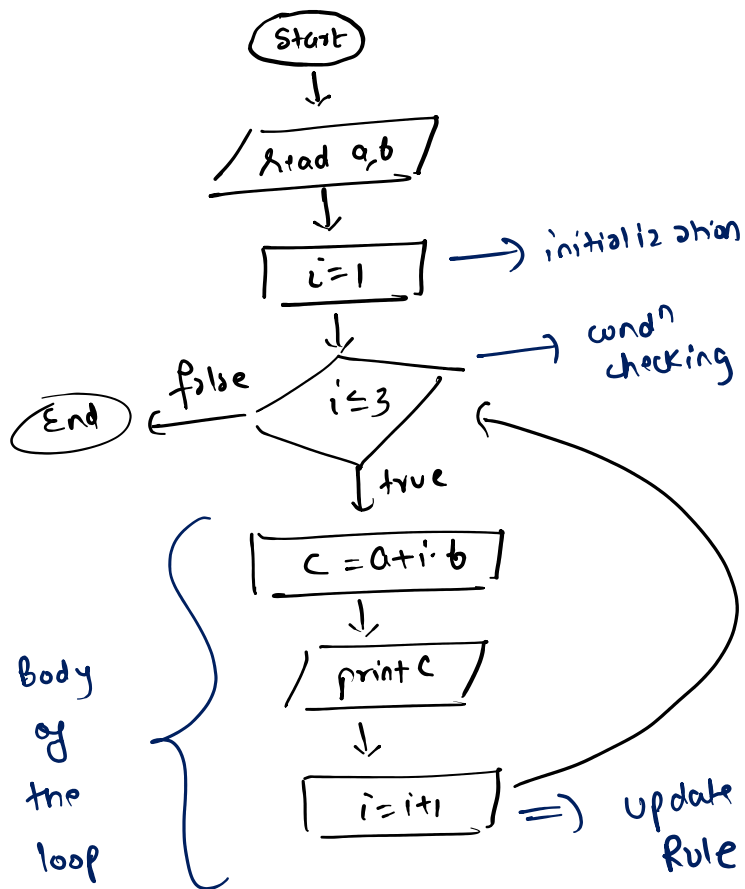
$$\begin{aligned} x &= a + b \\ y &= x + b \\ z &= y + b \end{aligned}$$

$\begin{matrix} \square & \square & \square & \square \\ 2 & 4 & 6 & 10 \\ a & b & i & c \end{matrix}$

$$\begin{aligned} 2 + 1 \cdot 4 &= 6 \\ 2 + 2 \cdot 4 &= 10 \\ 2 + 3 \cdot 4 &= 14 \\ 2 + 4 \cdot 4 &= 18 \end{aligned}$$



o/p: 6, 10, 14, 18, ...



4 3 2 10 14
~~2~~ ~~4~~ ~~14~~ ~~6~~
a b i c

1 <= 3? ✓
2 <= 3? ✓
3 <= 3? ✓
4 <= 3? X

2 + 1 * 4 = 6
2 + 2 * 4 = 10
2 + 3 * 4 = 14

o/p: 6, 10, 14

Loop

A **loop** is programming construct that allows you to do the **same task** repeatedly **while** a **condition is true**. In general, a loop has four components

- Initialization
- Condition Checking
- Body of the loop
- Update

Print Numbers

Given a positive number ' n ', design an algorithm to **print** the first n **natural numbers** & express it by a flowchart.

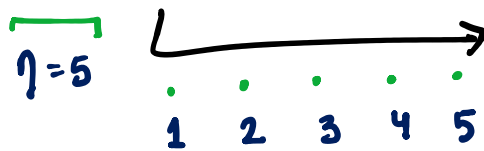
Example

Input : $n = 5$

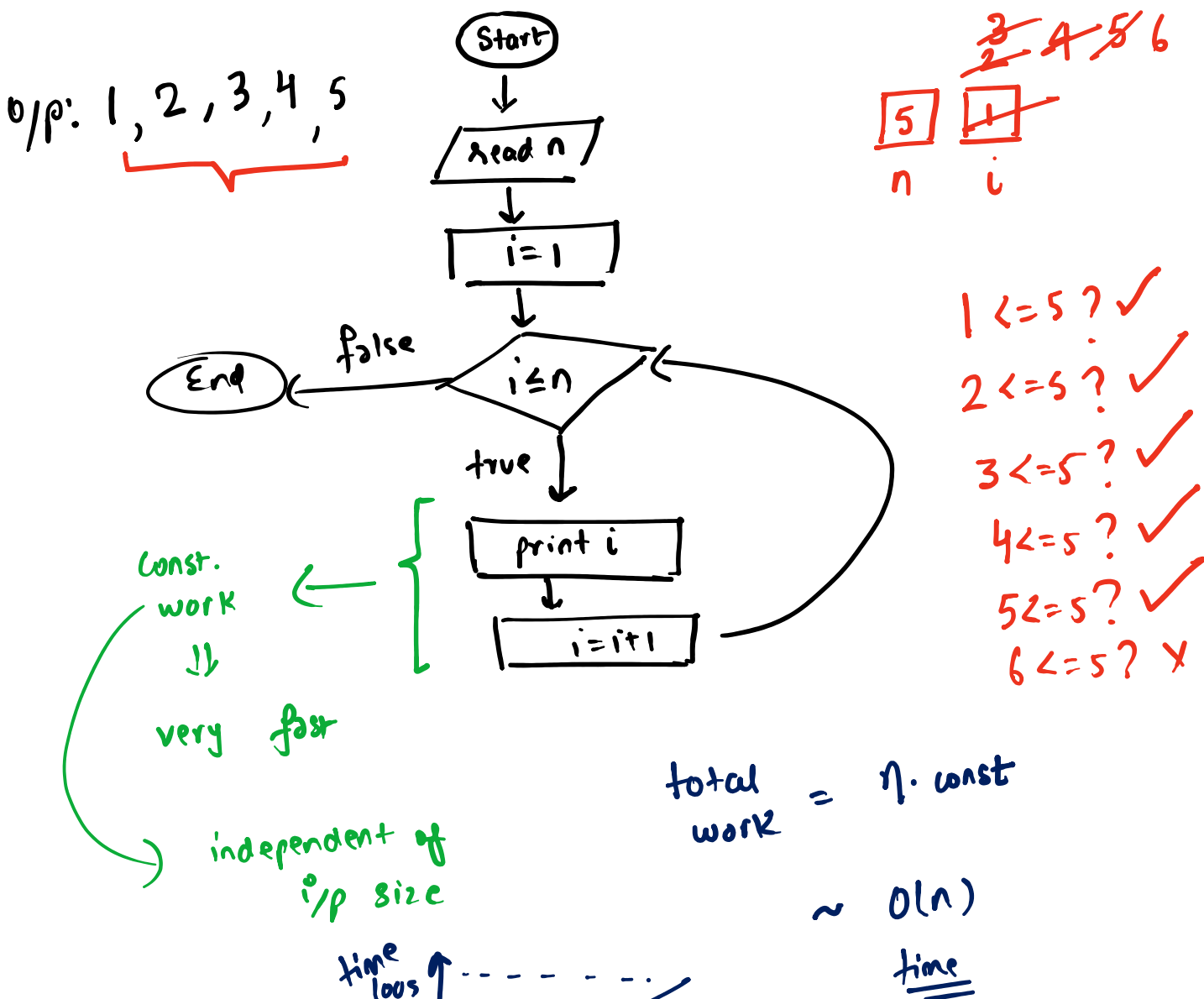
Output : 1, 2, 3, 4, 5

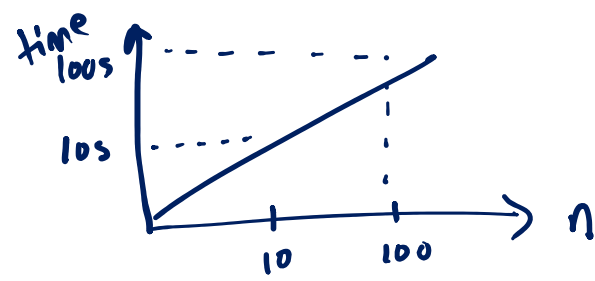
$n = 7$

1 2 3 4 5 6 7



iterate





time

Print Even Numbers

Given a positive number 'n', design an algorithm to **print** all the **even numbers** from **2 to n** & express it by a flowchart.

Example

Input : n = 8
Output : 2, 4, 6, 8

$n = 8$

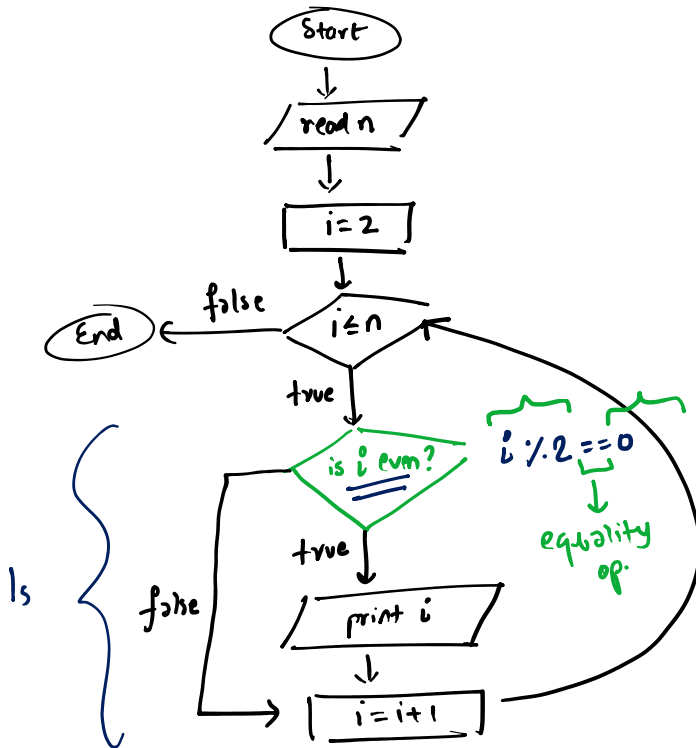
2	3	4	5	6	7	8
↑	·	↑	·	↑	·	↑
✓	x	✓	x	✓	x	✓

n i
6 12 3 4 5 6 7 o/p: 2, 4, 6

2 <= 6? ✓
3 <= 6? ✓
4 <= 6? ✓

5 <= 6? ✓
6 <= 6? ✓
7 <= 6? x

is 2 even? ✓
is 3 even? x
is 4 even? ✓
is 5 even? x
is 6 even? ✓



% => remainder operator

iterations = $n - 1 \sim n$

$a \% b = c$
when you div. a by b then remainder is c

work done in each iteration = const

$\therefore n \cdot c \sim O(n)$

i = 1
i = 2

i <= n
i <= n

i = i + 1
i = i + 1

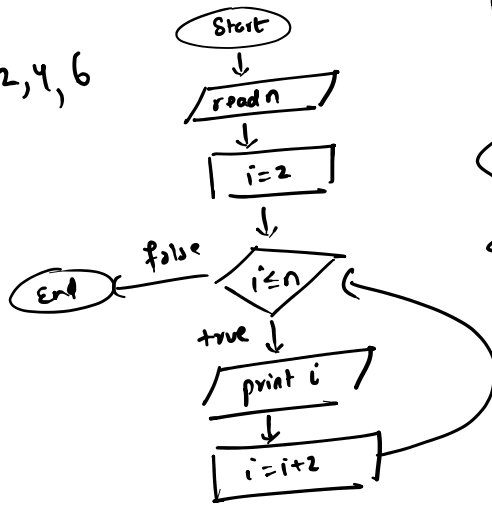
=> n times
=> n-1 times

10 % 3 = 1

17 % 4 = 1

19 % 5 = 4

o/p: 2, 4, 6



6 2 4 6 8
n i

2 ≤ 6? ✓
4 ≤ 6? ✓
6 ≤ 6? ✓
8 ≤ 6? ✗

iterations = $n/2$
work in each iteration = const
total work = $n/2 \cdot \text{const}$
 $\sim O(n)$

Sum of Natural Numbers

Given a positive number ' n ', design an algorithm to compute the **sum** of **first ' n ' natural numbers** & express it by a flowchart.

Example

Input : $n=5$

Output : 15

Explanation : $1+2+3+4+5 = 15$

$$\frac{n(n+1)}{2}$$

$$\frac{5(5+1)}{2} = 15$$

don't use
this formula
to solve this
question

$n=5$

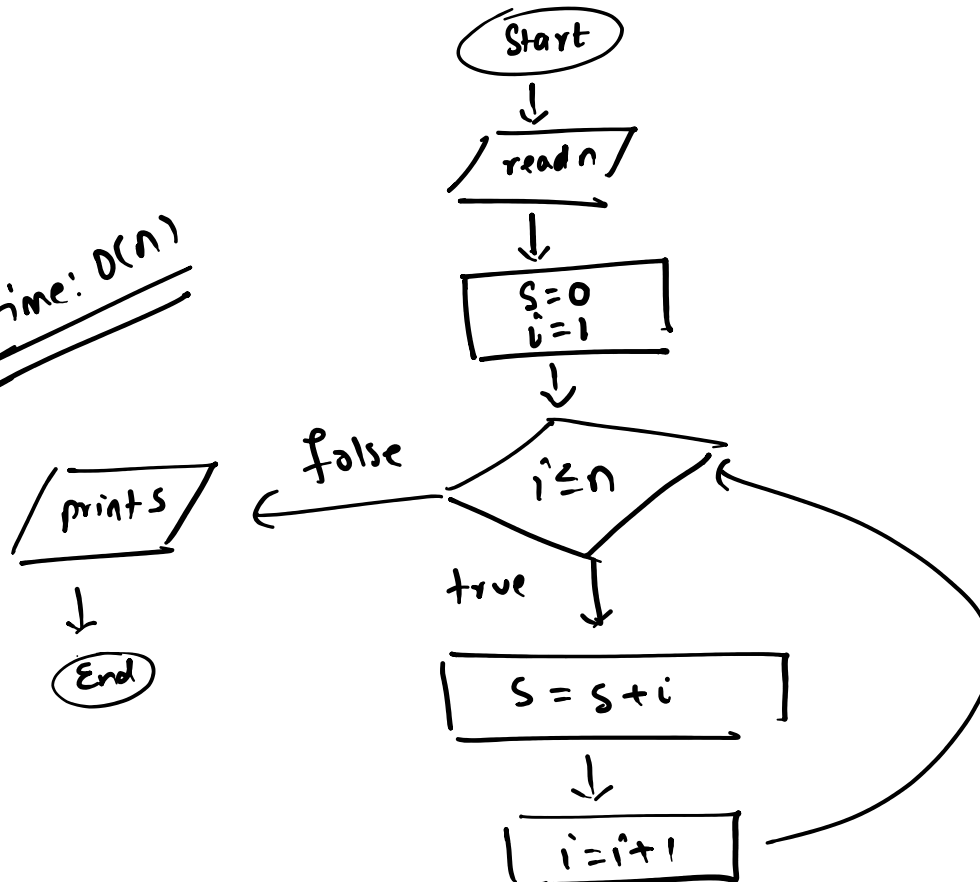
$$\begin{array}{ccccccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & + & 2 & + & 3 & + & 4 & + & 5 \end{array}$$

$$s = \cancel{0} + \cancel{1} + \cancel{2} + \cancel{3} + 15$$

$$\begin{array}{r} \boxed{15} \\ \hline 10 \\ \hline 5 \\ \hline 3 \\ \hline 2 \\ \hline 1 \end{array} \quad \begin{array}{r} 6 \\ \hline 5 \\ \hline 1 \\ \hline 0 \end{array}$$

$n \quad s \quad i$

time: $O(n)$



$$\begin{array}{l} 1 \leq 5? \checkmark \\ 2 \leq 5? \checkmark \\ 3 \leq 5? \checkmark \\ 4 \leq 5? \checkmark \\ 5 \leq 5? \checkmark \\ 6 \leq 5? \checkmark \end{array}$$

Sum of Numbers

Given a positive number ' n ' followed by ' n ' integer values design an algorithm to find the **sum** of ' n ' numbers & express it by a flowchart.

Example

Input : $n = 5$; [10, 20, 30, 40, 50]

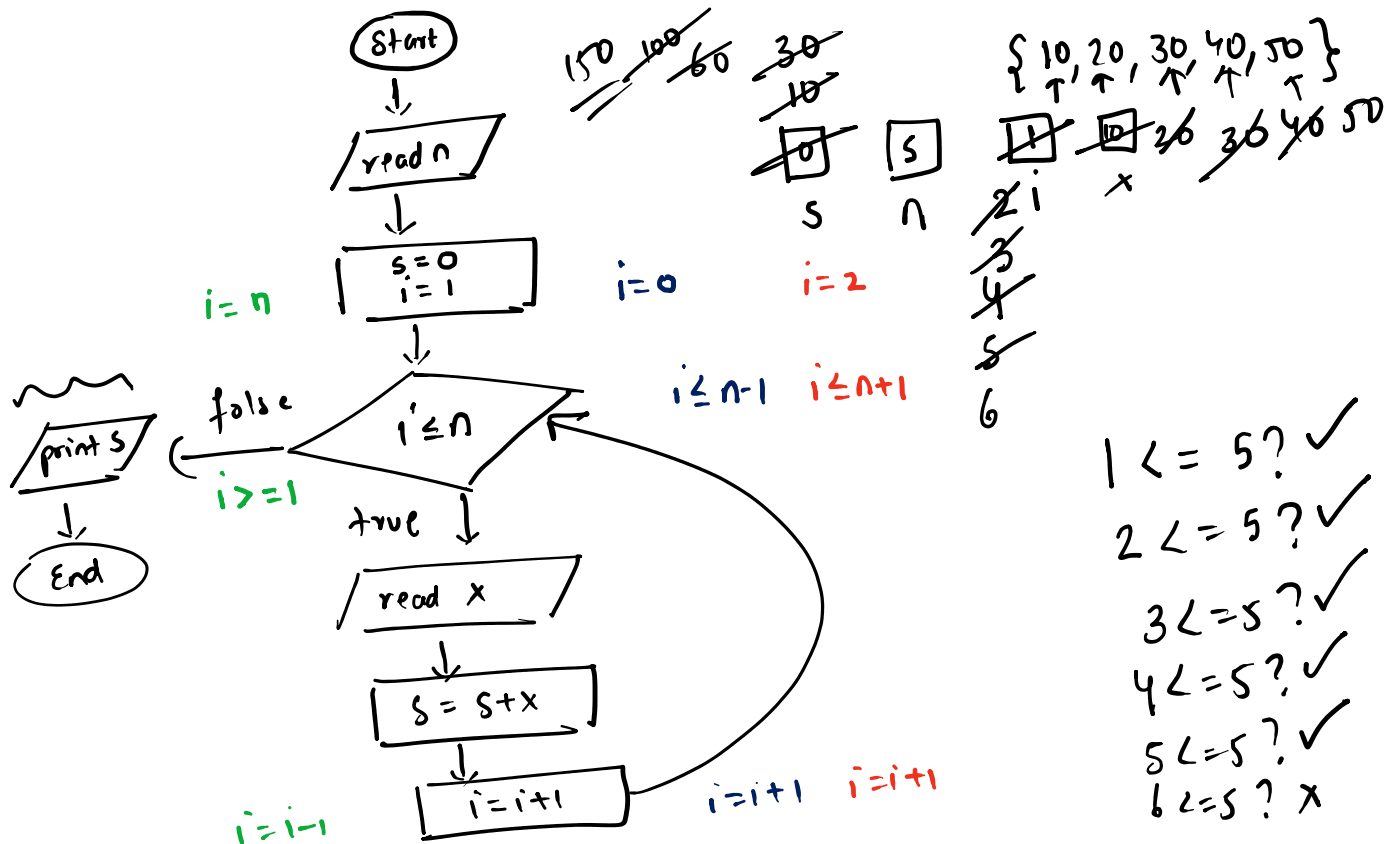
Output : 150

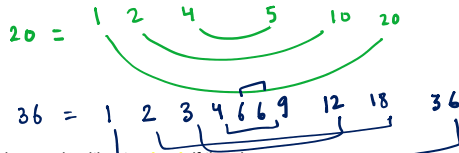
Explanation : $10+20+30+40+50 = 150$

$n=3$ (15, 10, 7)

o/p: 32

n





Check Prime

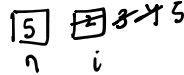
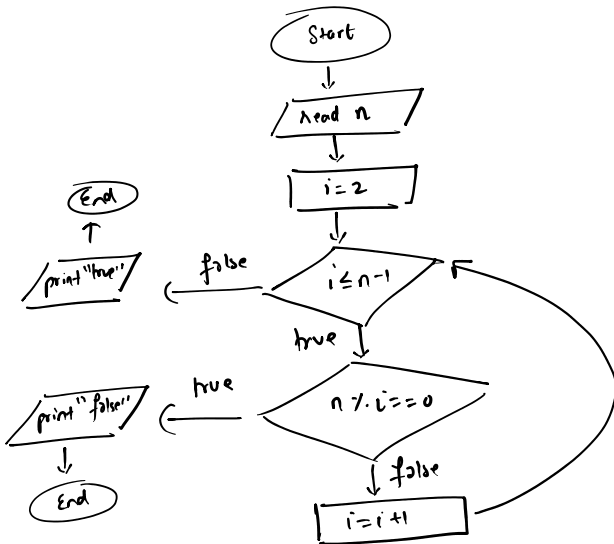
Given a positive number 'n' (>1), design an algorithm to check if the given number is a prime number or not & express it by a flowchart.

Example

Input : $n = 5$
Output : True

Input : $n = 4$
Output : False

A number n is a prime no. if it is divisible by only 1 & the number itself.



2 < 4? ✓
3 < 4? ✓
4 < 4? ✓
5 < 4? X

5 % 2 = 0? X
5 % 3 = 0? X
5 % 4 = 0? X

Worst-case

we'll hit the worst-case for this algo. when n is prime

#iterations = $n-2 \sim n$

work done in each it = const

∴ total time: $n \cdot c \sim O(n)$

$n=97$

2 to 96 ~ 100 it

⇒ time: $O(n)$

2 to $\sqrt{97} \sim 10$ it

$n=36$

$36 = 2 * 18$
 $= 4 * 9$
 $\vdots$

$n=20$

$20 = 4 * 5$

$n = a * b$

assume it is composite

assume $a \leq b$

it is
composite

=

$$\eta = 20$$

$$20 = 4 * 5$$
$$2 * 10$$
$$\vdots$$

$$\eta = a * b \leftarrow$$

$$a \leq b$$

mul. b/s by a

$$a^2 \leq ab$$

mul. b/s by b $\eta = 36$

$$ab \leq b^2$$

$$36 = 2 * 18$$
$$3 * 9$$
$$6 * 6$$
$$\vdots$$

$$a^2 \leq ab \leq b^2$$

$$a^2 \leq n \leq b^2$$

$$a \leq \sqrt{n} \leq b$$